A Formal Analysis of 5G ProSe AKA Protocols for U2N Relay Communication

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Abstract:
5G Proximity–based Service (ProSe) UE–to–Network (U2N) Relay can help a remote 5G User Equipment (UE) out of coverage connect with the network. The 3GPP committee has provided the standard Authentication and Key Agreement (AKA) protocols to achieve secure access for a Remote UE and establish a secure link between a Remote UE and a U2N Relay. However, the security of these AKA protocols is a remaining issue. At first, we present two detailed 5G ProSe AKA protocols over Control Plane (CP) and User Plane (UP) for U2N Relay communication referring to multiple related standards, then transform the security requirements for 5G ProSe U2N Relay communication as formal security properties, and provide two formal faithful security models for the 5G ProSe AKA protocols. We adopt the state–of–the–art formal verification tool Tamarin to achieve automated security analysis on two models through new proof strategies, and then find some significant and unexpected flaws. Finally, we propose corresponding measures that have least impact on standards based on the analysis on the attacks. Given that the version of Release 17 (R17) of the 3GPP standard has just been frozen, our work can provide a reference for the subsequent evolution of the protocols in 5G ProSe.

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